

Stephen Welch

Gainesville, VA

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Education

M.S. in Computer Engineering: Software & Machine Intelligence

May 2025 | GPA: 3.88

Relevant Courses: Learning Theory for Dynamics & Control | Machine Learning | Adaptive Control

Virginia Polytechnic Institute

Aug. 2023 - May 2025

B.S. in Computer Engineering: Controls, Robotics & Autonomy

Minor in Computer Science | GPA: 3.27

Virginia Polytechnic Institute

Aug. 2019 - May 2023

Skills

Tools Git, Gradle, JetBrains IDEs, Visual Studio, GNUPlot, $\LaTeX$
Frontend JavaFX, Swing, Qt, ASP.NET
Languages Python, Java 8, C++, C#, SQL

Work Experience

Software Engineer - Humanoid Controls

Apptronik

Austin, TX

Jun. 2024 - Present

- Built training, evaluation & inference pipeline to deploy the first reinforcement learning policies on the Apollo humanoid robot using **Isaac Lab** & **MuJoCo**
- Developed policies and sim-to-real pipelines for whole-body teleoperation, skill imitation from motion capture, and natural locomotion

Robotics Researcher

Terrestrial Robotics Engineering and Controls Lab

Blacksburg, VA

Jan. 2020 - Present

- Researched applications of convex optimization and reinforcement learning to robotics tasks
- Designed **Java** software enabling QP-based whole-body control of a 32 DoF parallel-actuated bipedal robot for balancing & locomotion tasks

Research Engineer

Occupational Ergonomics & Biomechanics Lab

Blacksburg, VA

Jan. 2024 - Dec. 2024

- Led development of UI & operational space controller for collaborative manufacturing studies with **C++**, **Qt**, and **ROS**

Machine Learning Intern

Shield AI

Alexandria, VA

Dec. 2020-May 2023

- Applied model predictive control with learned dynamics models to autonomous jets, enabling a 90% improvement in sample efficiency
- Developed optimization-based approaches to multi-vehicle coordination & target selection
- Built air-to-air combat simulation framework for large-scale (4v4+) dogfights in **PyTorch** & **RLlib** used across multiple government contracts

Enterprise Applications Intern

Serco North America

Herndon, VA

Mar. 2020 - Aug. 2020

- Wrote utilities to verify website integrity using **Java**, **JavaFX**, **Swing**, and **Selenium** with interactive graph visualizations
- Performed bugfixes and quality-of-life changes on widely used TA/NDA request web application built using **C#**, **ASP.NET** and a **SQL** database
- Eliminated time-consuming manual labor for Contracts department using **PowerShell** scripts to automatically update macro-enabled spreadsheets

Projects

AUV Fault Detection

Center for Marine Autonomy and Robotics

2023

- Integrates with IMU-based state estimator to detect actuator & sensor faults on the 690-type submersible robot, increasing reliability during long-term surveys of the oceanbed
- Implemented & validated in **C++ ROS**-based simulation

Miniature Bipedal Walker

Personal Project

In Progress

- Designing and building torque-controlled 3 degree-of-freedom parallel-actuator bipedal robot as a <\$1000 learning platform for motion planning & control

Multi-Agent Heron Manager

Heron Systems

2021

- **Python** framework enabling multi-agent interaction with multi-threaded and distributed compute capability
- Designed in collaboration with senior engineers and developed with 2 fellow interns
- Integrates with open-source and in-house reinforcement learning libraries
- Implements multi-agent air-to-air combat environment using JSBSim and RLLib
- Actively maintained in-house

Robot Arm

Personal Project

2019

- Designed and 3D printed low-cost 2 degree-of-freedom robot arm using SOLIDWORKS
- Wrote **C++** Arduino code receive commands from PC over serial and drive stepper motors
- Created **Qt** graphical interface in **C++** to control arm over serial
- Implemented inverse kinematics algorithm for arm

Accomplishments

2018-2019	Advanced Computer Studies Student of the Year , Battlefield High School	Haymarket, VA
2018-2019	Dean's List , Northern Virginia Community College	Haymarket, VA
2017-2019	National Semi-Finalist , CyberPatriot	Haymarket, VA
Apr. 2018	Presenter for Effective Student Leadership in the FIRST Robotics Competition , FRC World Championship	Detroit, MI
Apr. 2022	Author for A Mapping Approach to Achieve Torque Control for Parallel-Actuated Robotic Systems , ASME 2022	TBD